



Conference and Labs of the Evaluation Forum (CLEF) 2026

From Words to Wit: Humor-Aware Information Retrieval in English and Hinglish

Arjun Mukherjee, Krishna Tewari, Jasvinder Singh, Sukomal Pal

Information Retrieval Lab (IReL)
Department of Computer Science and Engineering
Indian Institute of Technology (BHU) Varanasi, India

1. Task on CLEF 2026 JOKER

1.1 Task Description

1.2 Dataset

2. Methodology

2.1 Retrieval Pipeline

2.2 Humor-Aware Re-ranking

2.3 System Comparison

3. Results

3.1 Official Results

3.2 Error Analysis

3.3 Hinglish Subtask

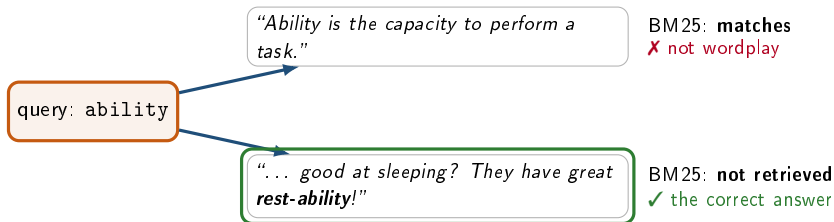
4. Conclusion

Task 1 — Humor-Aware Information Retrieval (HAIR)

Retrieve **short humorous texts** from a document collection for a given query — fulfilling the **dual criteria**: **relevant to the query and an instance of wordplay**.

Task 1 — Humor-Aware Information Retrieval (HAIR)

Retrieve **short humorous texts** from a document collection for a given query — fulfilling the **dual criteria**: **relevant to the query** and an **instance of wordplay**.



“Why did the musician get an A+ in alphabet class? Because they really knew their A!”

✓ wordplay, query a

“I lost my job at the calendar factory. Apparently, I didn’t have the ability to work dates!”

✓ wordplay, query ability

“Ability is the capacity to perform a task.”

✗ a typical distractor

Figure 1: Example documents: two instances of wordplay and a typical distractor.

	English	Hinglish
Documents	96,603	32,652
Train queries	10	10
Test queries	972	485
Train relevant docs	25	60

Q–A jokes, Tom Swifties and puns are mixed with **Wikipedia and LLM-generated distractors**. The first example is query a, which **both systems miss**.

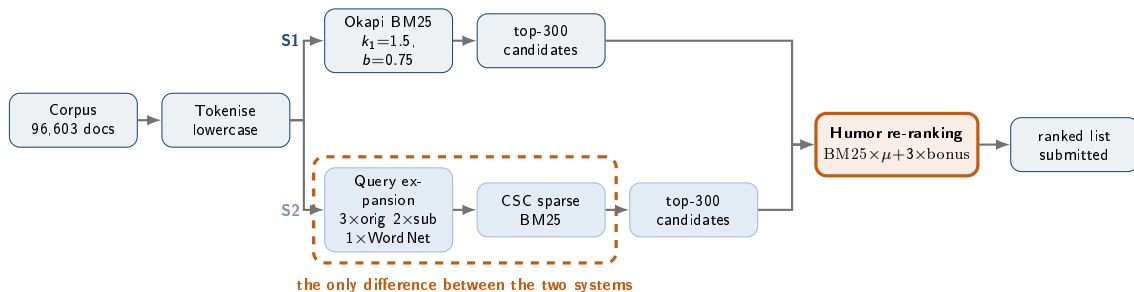


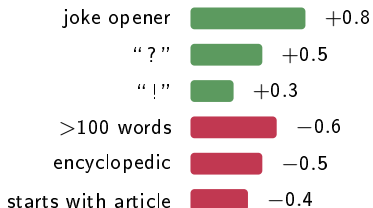
Figure 2: System 2 adds query expansion; both systems share the same humor re-ranking.

k_1 , b = BM25 saturation and length-normalisation parameters; **CSC** = compressed sparse column. Both systems are **unsupervised** — no GPU, no training, no labels — and run the full query set in **39.4 s** (S1) and **12.3 s** (S2).

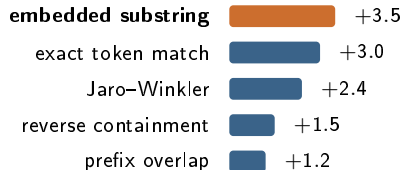
$$\text{score}(q, D) = \text{BM25}(q, D) \times \mu(D) + \text{bonus}(q, D) \times 3.0$$

q = query, D = document; $\mu(D)$ = **humor multiplier** (does D look like a joke?); $\text{bonus}(q, D)$ = **wordplay bonus** (is D a pun on q ?), where **embedded substring** means q occurs inside a longer token and **Jaro-Winkler** is a phonetic string similarity in $[0, 1]$.

$\mu(D)$ — joke-shaped?



bonus — a pun on q ?



	System 1 (BM25 + humor)	System 2 (QE + BM25)
Query	used as written	expanded : $3\times$ original, $2\times$ substring (top 8), $1\times$ WordNet (max 5)
Index	rank_bm25 Okapi index	CSC sparse-matrix BM25
Candidates	top 300	top 300
Re-ranking	$\text{BM25} \times \mu + 3 \times \text{bonus}$	identical
Runtime	39.4 s, full run	12.3 s , full run
Strength	noise-free signal; best precision	recall on distinctive pun terms
Weakness	misses morphological variants	over-expansion noise on short queries

Table 1: Side-by-side comparison of System 1 and System 2.

The two differ in **exactly one stage** — query expansion.

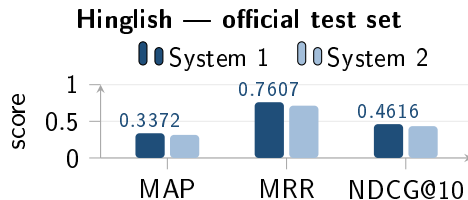
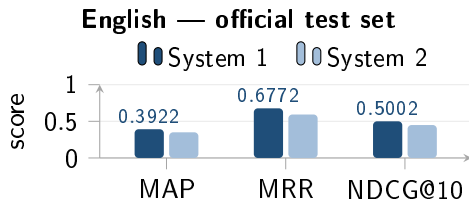


Figure 3: Official test-set scores for both subtasks.

System 1 scores higher on **every metric shown**; the **largest absolute gap is in MRR**, as expansion noise displaces the top hit. Yet both systems fail entirely on some queries:

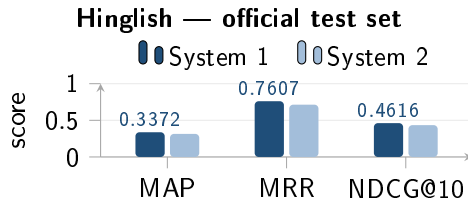
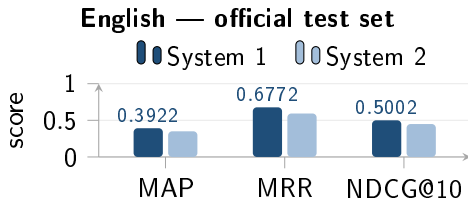


Figure 3: Official test-set scores for both subtasks.

System 1 scores higher on **every metric shown**; the **largest absolute gap is in MRR**, as expansion noise displaces the top hit. Yet both systems fail entirely on some queries:

Stop-word pivot

a: 7 judgements, none retrieved — no BM25 signal.

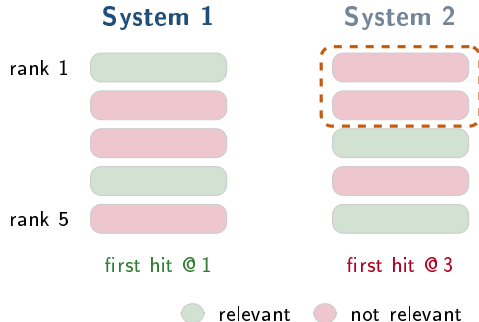
Morphological gap

acidic → acidity, acidly: captured by neither system.

Transliteration

Movie → muvi, filum: spellings a token index cannot match.

schematic



- Queries are **single words**, and short queries amplify noisy expansions.
- Recall gains are local — abrupt improves, yet System 2 retrieves **fewer** relevant documents overall (2,544 vs. 2,689, English).
- On Hinglish the same loss, for a different reason: substring and WordNet rules encode **English morphology**.
- High MRR with low MAP is structural — $\mu(D)$ is **query-independent**, so once every candidate looks joke-shaped it cannot discriminate further.

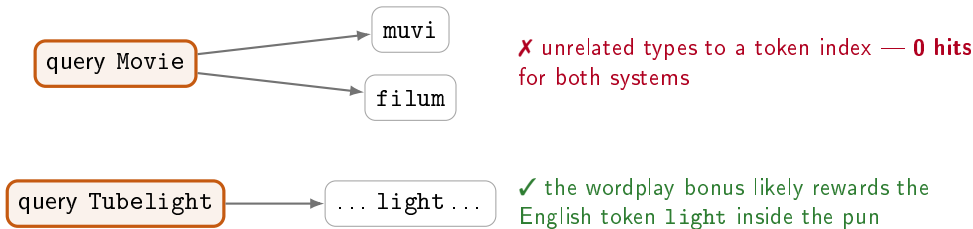


Figure 4: Two Hinglish training queries: Movie and Tubelight.

- Code-mixed puns turn on **cross-lingual phonetic overlap**, and one pivot may be spelled several ways — `filum` for `film` — so a token index never sees them as the same word.
- Fixing this needs a **character-level index** or **phonetic normalisation**, neither of which an English corpus requires. Jaro–Winkler partially bridges the gap: it handles `Chhat` and `Da1`, yet `Movie` still returns zero.

hits@20, 6 relevant per query

Query	S1	S2
Tubelight	3	2
Movie	0	0
other 8 queries	identical	

Table 2: hits@20 as confirmed
in the paper's text.

- **Identical on 9 of 10 queries** — direct evidence that the shared BM25 backbone is genuinely language-agnostic, and that English-tuned expansion adds nothing here.
- **Tubelight** is the single exception: System 1 finds one more, likely because the wordplay bonus rewards the English token *light* inside the pun.
- **Movie**: both return zero — the relevant jokes spell it *muvi* or *filum*.
- Chhat, Dal: Jaro-Winkler **partially bridges** the cross-lingual phonetic gap — a dedicated Hinglish phonetic model would do better.

- **Main Contributions:**

- Two unsupervised, CPU-only systems adding a rule-based humor multiplier and a wordplay bonus to sparse BM25.
- To our knowledge, the first BM25 system with humor-aware re-ranking for Hinglish pun retrieval.

- **Findings:**

- **System 1** scores higher than System 2 on MAP, MRR and NDCG on both official test sets (English MAP 0.3922 / MRR 0.6772; Hinglish MAP 0.3372 / MRR 0.7607).
- Query expansion adds noise on short or ambiguous queries, and helps only where the pun term is morphologically distinctive.
- Limits: no plain-BM25 ablation and hand-set weights; 58.8% (EN) and 53.0% (HI) of relevant documents are retrieved.

- **Future Work:**

- Character n -gram BM25 indices, a dedicated Hinglish humor classifier, ColBERT-style dense re-ranking, and adaptive expansion weighting.

- [1] L. Ermakova et al. *Overview of CLEF 2026 JOKER track*. CLEF 2026, Springer LNCS.
- [2] P. Vachharajani et al. *CLEF 2026 JOKER Task 1: humor-aware IR in English and Hinglish*. CLEF 2026 Working Notes.
- [3] L. Ermakova, A.-G. Bosser, A. Jatowt, T. Miller. *The JOKER corpus: English–French parallel data for wordplay recognition*. SIGIR 2023.
- [4] Y. Xie, J. Li, P. Pu. *Uncertainty and surprisal jointly deliver the punchline*. ACL-IJCNLP 2021.
- [5] T. Chaudhari, A. Vora, S. Hotha, S. Sonawane. *PICT at CLEF 2025 JOKER track: BERT-enhanced ensembles*. CLEF 2025 Working Notes.

Queries?

Thank You!

Team IReL_IIT(BHU) — IIT (BHU) Varanasi, India

Official test results

English test			Hinglish test		
Metric	S1	S2	Metric	S1	S2
MAP	.3922	.3500	MAP	.3372	.3148
MRR	.6772	.5929	GMAP	.2119	.1745
P@10	.2262	.2099	MRR	.7607	.7135
NDCG@5	.4733	.4156	R-prec	.3474	.3271
NDCG@10	.5002	.4493	NDCG@10	.4616	.4342
NDCG@100	.5424	.4944	NDCG@20	.5091	.4840
Rel. retr.	2689	2544	Rel. retr.	1542	1501

GMAP = geometric mean average precision; R-prec = precision at the number of relevant documents.

Hyper-parameters

BM25 k_1 / b	1.5 / 0.75
Candidate pool	top 300
Filtered vocabulary	36,015 tokens
TF matrix non-zeros	1,602,774
Expansion weights	$3 \times / 2 \times / 1 \times$
Substring candidates	top 8 by freq.
WordNet lemmas	max 5
μ : ?/!/quotes	+0.5/+0.3/+0.2
μ : joke opener	+0.8
μ : >100 words / article / encycl.	-0.6/-0.4/-0.5
μ : floor	≥ 0.05
Bonus: embedded substring	+3.5
Bonus: exact token match	+3.0
Bonus: Jaro-Winkler	up to +2.4
Bonus: reverse cont. / prefix	+1.5 / +1.2